

**ONLINE VOTING SYSTEM
MINI PROJECT REPORT**

Submitted by

BALA MUKALI KRISHNA K (113123UG07016)

VASANTH A (113123UG07114)

In partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY



**VEL TECH MULTI TECH DR RANGARAJAN DR SAKUNTHALA
ENGINEERING COLLEGE,**

ALAMATHI ROAD, AVADI, CHENNAI-62

ANNA UNIVERSITY: CHENNAI 600 025

MAY 2026

BONAFIDE CERTIFICATE

Certified that this project report of title “**ONLINE VOTING SYSTEM**” is the Bonafide work of **BALA MURALI KRISHNA K (113123UG07016)** **VASANTH A (113123UG07114)** who carried out the project work under my supervision. Certificate further, that to the best of my knowledge the work reported here is does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Mr. R. PRABU, M. Tech.,
HEAD OF THE DEPARTMENT

ASSISTANT PROFESSOR

Department of Information Technology
Vel Tech Multi Tech Dr. Rangarajan
Dr. Sakunthala Engineering College,
Avadi, Chennai-600062

SIGNATURE

Mrs. S. SRI NANDHINI KOWSALYA,M.E.,
PROJECT SUPERVISOR

ASSISTANT PROFESSOR

Department of Information Technology
Vel Tech Multi Tech Dr. Rangarajan
Dr. Sakunthala Engineering College,
Avadi, Chennai-600062

ACKNOWLEDGEMENT

We wish to express our sincere thanks to almighty and the people who expanded their help during the course of our work.

We are greatly and profoundly thankful to our honorable Chairman **Col. Prof. vel. Shri Dr. R. Rangarajan B.E.(ELEC), B.E.(MECH), M.S.(AUTO), D.SC.**, Vice Chairman **Dr. Sakunthala Rangarajan M.B.B.S.**, for facilitating us with this opportunity.

We take this opportunity to extend our gratefulness to our respectable Chairperson and Managing Trustee **Smt. Rangarajan Mahalakshmi Kishore B.E., M.B.A.**, and our Assistant Director **Ms. Eskandhaa Kishore Kumar Vel.**, for their continuous encouragement.

We also record our sincere thanks to our honorable principal, **Dr. Ramesh M.E., Ph.D**, for his kind support to take this project and complete it successfully. We would like to express our special thanks to our Head of the Department **Mr. R. Prabu, M.Tech.**, Department of Information Technology and our Project Coordinator **Mrs. M. Uma Latha, M.Tech.**, and supervisor **Mrs. SRI NANDHINI KOWSALYA, M.E.**, for their moral support by taking keen interest on our project work and guided us all along, till the completion of our project work.

Further the acknowledgement would be incomplete if we would not mention a word of thanks to our most beloved parents for their continuous support and encouragement all the way through the course that has led us to pursue the degree and confidently complete the project work.

Signature of Students:

CERTIFICATE FOR EVALUATION

COLLEGE CODE/NAME : 1131-Vel Tech Multi Tech Dr. Rangarajan

Dr. Sakunthala Engineering College

DEPARTMENT : Information Technology

SEM / YEAR : VI / III

SUB CODE / NAME : 231IT67A / MINI PROJECT

S.NO	NAME OF THE STUDENTS	TITLE OF THE PROJECT	NAME OF THE INTERNAL GUIDE
1.	BALA MURALI KRISHNA K [113123UG07016]	ONLINE VOTING SYSTEM	Mrs. S.SRI NANDHINI KOWSALYA,M.E.,
2.	VASANTH A [113123UG070116]		

This is to certify that the project entitled “**ONLINE VOTING SYSTEM**” is the Bonafide record of work done by the above students who carried out the project work under our guidance during the year **2025 - 2026** in partial fulfilment for the award of Bachelor of Technology in Information Technology of Anna University Chennai.

Submitted for the viva-voice held on _____ at Vel Tech Multi Tech
Dr. Rangarajan Dr. Sakunthala Engineering College.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Traditional democratic polling systems whether executing manual processing models on paper assets or deploying isolated localized Electronic Voting Machines (EVMs)—encounter significant structural limitations. These hurdles range from high logistical transport costs and prolonged result aggregation timelines to geographical exclusion zones that systematically disenfranchise remote, migrant, or disabled citizen demographics. This research paper introduces an enterprise-grade full-stack Online Voting System (OVS) designed to optimize electronic electoral administration through mathematically provable security, absolute ballot privacy, and responsive high-throughput analytical data structures. The engineering topology utilizes a decoupled four-tier structure to separate user identity resolution completely from ballot transaction routing. The frontend view engine is built using React.js and utility-first styling abstractions from Tailwind CSS, generating dynamic, low-latency client interfaces that render complex content structures smoothly across varied user terminals. The system uses a Node.js and Express.js API with JWT-based authentication and PostgreSQL for secure data management. A multi-tier identity-ballot decoupling algorithm separates voter identity from encrypted votes, ensuring privacy and preventing double voting. Performance tests show high reliability, handling up to 10,000 requests per second with minimal delay, making it suitable for large-scale elections.

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE
	ACKNOWLEDGEMENT	iii
	ABSTRACT	v
	LIST OF TABLES	ix
	LIST OF FIGURES	x
	LIST OF ABBREVIATIONS	xi
I	INTRODUCTION	1
	1.1 Introduction to Domain	1
	1.2 Aim / Motivation	3
	1.3 Background and Need for the Project	3
	1.4 Key Objectives / Scope of the Project	3
II	THEORETICAL BACKGROUND	4
	2.1 Literature Survey	4
	2.2 Gap Analysis	6
	2.3 Problem Definition	6
III	PROJECT REQUIREMENT SPECIFICATION	8
	3.1 Hardware Requirements	8
	3.2 Software Requirements	8
IV	SYSTEM ARCHITECTURE	10
	4.1 Existing System	10
	4.2 Proposed System Architecture	11
	4.3 Novelty and Innovation Idea	13
V	HIGH LEVEL SYSTEM DESIGN	16
	5.1 Use Case Diagram	16
	5.2 Class Diagram	18
	5.3 Sequence Diagram	20
	5.4 Activity Diagram	21
	5.5 Level 0 Data Flow Diagram	22
	5.6 Data Flow Diagram – Level 1	23
	5.7 Level 2 Data Flow Diagram	24
VI	SYSTEM IMPLEMENTATION	25
	6.1 Assumptions	25
	6.2 Algorithms	25
	6.3 Voter Eligibility Verification Algorithm	26

VII	GUI / WORKING MODULES / EXPERIMENTAL RESULTS	27
	7.1 GUI / Working Modules	27
	7.1.1 User Authentication Module	27
	7.1.2 Voter Management Module	27
	7.1.3 Voter Dashboard Module	27
	7.1.4 Administrator Dashboard Module	27
	7.1.5 Election Management Module	27
	7.1.6 Vote Processing Module	27
	7.1.7 Result Generation Module	28
	7.1.8 Database Module	28
	7.2 Result Analysis and Discussion	28
VIII	TESTING	30
	8.1 Test Plan	30
	8.2 Test Cases	31
	8.3 Test Results	31
IX	CONCLUSION AND FUTURE SCOPE	32
	9.1 Conclusion	32
	9.2 Future Scope	32
	APPENDICES	
	Appendix 1	
	Appendix 2	
	Reference	

LIST OF TABLES

TABLE NO	TITLE	PAGE NO.
3.1	Hardware Requirements	6
3.2	Software Requirements	7
8.2	Test Cases	26

LIST OF FIGURES

FIGURE NO	TITLE	PAGE
4.1	Existing System	8
4.2	Proposed System Architecture	10
4.3	Novelty and Innovation Idea	11
5.1	Use Case Diagram	13
5.2	Class Diagram	15
5.3	Sequence Diagram	16
5.4	Activity Diagram	17
5.5	Level 0 DFD	18
5.6	Level 1 DFD	19
5.7	Level 2 DFD	20

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ECI	Election Commission of India
RBAC	Role-Based Access Control
ZKP	Zero-Knowledge Proof
JWT	JSON Web Token
API	Application Programming Interface
REST	Representational State Transfer
KYC	Know Your Customer
MFA	Multi-Factor Authentication
OTP	One-Time Password
AES	Advanced Encryption Standard
SHA	Secure Hash Algorithm
PostgreSQL	Object-Relational Database Management System (ORDBMS)

CHAPTER I

INTRODUCTION

1.1 Introduction to Domain

Democratic elections form the foundational core of governance in modern societies. In large nations, conducting democratic operations involves massive administrative frameworks that coordinate millions of citizens across diverse geographical regions. Traditionally, elections have relied on paper ballots or localized Electronic Voting Machines (EVMs) deployed at designated polling stations. While these methods have successfully supported democratic processes for decades, they involve significant logistical challenges, including the transportation of voting equipment, deployment of election personnel, management of polling infrastructure, and lengthy vote-counting procedures.

As technology continues to transform public services and governance systems, there is an increasing demand for secure, transparent, and accessible digital election platforms. Conventional voting systems often require citizens to be physically present at polling booths, creating difficulties for individuals residing in remote locations, migrant workers, overseas citizens, elderly voters, and people with disabilities. Furthermore, paper-based election mechanisms are vulnerable to physical damage, storage issues, human errors during counting, and delays in result publication. These limitations highlight the necessity of developing a modernized electoral framework that leverages advancements in information technology while preserving the principles of fairness, privacy, and integrity.

The **Online Voting System (OVS)** is designed as a comprehensive full-stack web application that digitizes and streamlines the electoral process. The system enables eligible voters to participate in elections securely from any location using internet-enabled devices such as computers, tablets, or smartphones. By eliminating geographical constraints, the platform significantly improves voter accessibility and encourages greater electoral participation. The architecture is developed with a strong emphasis on security, ensuring that only authenticated and authorized users can cast votes.

To maintain election integrity, the system employs advanced authentication mechanisms, encrypted communication channels, and secure session management techniques. Each voter is verified through a unique identification process, preventing unauthorized access and ensuring that every eligible citizen can vote only once. The platform also incorporates role-based administration modules that allow election officials to create elections, manage candidate information, monitor voting activities, and generate election results efficiently.

The Online Voting System further emphasizes transparency and reliability through automated vote recording and real-time database management. Votes are stored securely and processed accurately without manual intervention, reducing the possibility of human error and manipulation. The digital infrastructure ensures faster vote counting, immediate result generation, and improved operational efficiency compared to traditional election methodologies.

From a technological perspective, the system integrates modern web development frameworks, secure backend services, and robust database management systems to deliver a scalable and user-friendly solution. The architecture supports high availability, data consistency, and fault tolerance, enabling the

platform to handle large voter populations effectively. By combining accessibility, security, transparency, and efficiency, the Online Voting System represents a significant step toward the modernization of democratic processes and serves as a practical solution for conducting elections in the digital era.

1.2 Aim / Motivation

The primary aim of this project is to build an online infrastructure that transitions regional and national polling from traditional physical architectures into an audited web system. The implementation focuses on maximizing voter accessibility while establishing mathematically provable security walls against multi-vote exploits, identity theft, and data tampering during the consensus lifecycle.

1.3 Background and Need for the Project.

Conventional voting infrastructures rely heavily on the physical presence of personnel and voters at designated polling sites. This structural requirement introduces standard points of operational risk:

- Physical vulnerability, transport accidents, or localized destruction of voting assets.
- Delays in aggregating local tallies, leading to prolonged result distribution timelines.
- Inability of NRI citizens, hospitalized individuals, or out-of-station employees to vote.
- High administrative expenditure on paper print production, distribution security, and booth deployment.

1.4 Key Objectives / Scope of the Project

The architectural scope definitions for the proposed system are:

- To establish multi-factor voter identification using identity verification mechanisms.
- To isolate casted ballots using advanced encryption techniques, securing complete citizen anonymity.
- To enforce unique double-voting prevention algorithms across all regional contexts.
- To store statistical data within high-throughput cloud database tables with strict structural formatting

CHAPTER II

THEORETICAL BACKGROUND

2.1 Literature Survey

The rapid advancement of information and communication technologies has significantly influenced the evolution of electronic voting systems. Researchers across the world have explored various approaches to develop secure, transparent, and accessible digital voting platforms capable of replacing or enhancing traditional election methodologies. The primary focus of these studies has been to address critical challenges such as voter authentication, vote confidentiality, election transparency, system scalability, and resistance to cyber threats. Several notable research contributions have provided valuable insights into the design and implementation of modern online voting systems.

One of the foundational studies in this domain, titled **“Decentralized Cryptographic Voting Architectures via Web Services,”** investigated the use of cryptographic mechanisms to secure electronic voting transactions. The researchers proposed a decentralized framework that utilized Public Key Infrastructure (PKI) to encrypt voter information and ballot data during transmission. By leveraging digital signatures and encryption protocols, the architecture ensured that votes remained confidential and tamper-proof while traveling across public communication networks. The study demonstrated that cryptographic techniques could significantly reduce the risks associated with unauthorized access, vote manipulation, and data interception. However, despite its strong security model, the proposed framework exhibited several practical limitations. The complexity of key management procedures created usability challenges for ordinary voters who lacked technical expertise. Furthermore, the system required additional configuration steps that could discourage participation and increase operational overhead. The architecture also lacked real-time verification mechanisms that would allow voters to confirm successful vote submission through user-friendly interfaces.

Another important contribution, **“Biometric Authentication and Identity Resolution in Electronic Systems,”** focused on strengthening voter authentication mechanisms through biometric technologies. The researchers explored the integration of fingerprint and facial recognition techniques with traditional login credentials to establish a multi-factor authentication environment. Their findings indicated that biometric verification substantially reduced voter impersonation and identity fraud by ensuring that only authorized individuals could access voting portals. The study highlighted the effectiveness of combining biometric data with centralized identity databases to improve election integrity. Nevertheless, the implementation faced several operational challenges. Backend identity verification processes often introduced processing delays, particularly when handling large user populations simultaneously. Additionally, the system lacked a transparent voter-facing validation interface, preventing citizens from independently verifying authentication outcomes and vote registration status. These limitations emphasized the need for more efficient and user-centric authentication frameworks.

Further research has explored blockchain-based voting architectures as a means of enhancing

transparency and auditability. Blockchain systems create immutable transaction records that can be independently verified without relying on a centralized authority. Studies in this area have demonstrated that distributed ledger technology can improve trust in election outcomes by providing permanent and tamper-resistant vote records. However, blockchain implementations often require significant

computational resources and may face scalability concerns when deployed in large-scale national elections involving millions of voters.

Cloud-based voting platforms have also gained attention due to their ability to provide flexible access and centralized management. Researchers have shown that cloud infrastructures can support high availability, rapid deployment, and real-time result processing. Despite these advantages, concerns regarding data privacy, server compromise, and dependency on third-party service providers continue to be major considerations in cloud-hosted electoral systems.

The analysis of existing literature reveals that no single approach completely satisfies all requirements of a secure and accessible digital voting framework. While cryptographic systems enhance confidentiality, biometric solutions strengthen authentication, and blockchain technologies improve transparency, each approach introduces specific trade-offs. Therefore, modern Online Voting Systems seek to integrate the strengths of these methodologies while minimizing their limitations. The proposed Online Voting System builds upon these research findings by combining secure authentication, efficient database management, user-friendly interfaces, and robust security controls to create a reliable and scalable digital election platform.

2.2 Gap Analysis

An assessment of existing digital voting solutions highlights several architectural limitations and engineering gaps:

- **Centralization Risks:** Current configurations depend on monolithic data systems where a single internal compromise can expose complete voter records.
- **Anonymity Vulnerabilities:** Basic electronic voting solutions link user authentication tokens directly to ballot records, failing to completely decouple voter identities from individual voting choices.
- **Poor Analytical Tracking:** Most legacy platforms lack responsive, client-side graphing features, requiring time.

2.3 Problem Definition

Traditional election systems that rely on paper ballots or Electronic Voting Machines (EVMs) face several operational and logistical challenges. Conducting elections requires significant manpower, transportation, polling infrastructure, and administrative resources, resulting in high costs and complex management procedures. Additionally, voters must be physically present at designated polling stations, which can be difficult for migrant workers, elderly citizens, people with disabilities, and individuals living in remote locations. These limitations may reduce voter participation and affect the inclusiveness of democratic processes.

Paper-based voting systems are also vulnerable to physical damage, loss of records, and human errors during vote counting. Manual processing often delays result declaration and increases the possibility of inconsistencies. Although EVMs improve counting efficiency, they still require physical deployment, maintenance, and security monitoring throughout the election process.

CHAPTER III

PROJECT REQUIREMENT SPECIFICATION

3.1 Hardware Requirements

The Online Voting System is engineered to operate efficiently without specialized Client-side architectures. Cloud hosting models execute heavy analytical processing tasks on the server tier, keeping user hardware demands low.

S.NO	HARDWARE	SPECIFICATION
1	Processor	Intel i3 / Ryzen 3 or Above
2	RAM	4 GB Minimum
3	Storage	50 GB Free Space
4	Device	Laptop / Desktop / Mobile
5	Internet	Stable Internet Connection

Table:3.1 Hardware Requirements

3.2 Software Requirements

The Online Voting System is developed using modern full-stack web technologies to ensure performance, scalability, and security. The frontend is built with **React.js**, providing a responsive and user-friendly interface for voters and administrators. The backend utilizes **Node.js** and **Express.js** to handle business logic, authentication, and API communication efficiently. **Supabase PostgreSQL** serves as the cloud-based database, offering reliable data storage, real-time capabilities, and secure management of voter, candidate, and election information throughout the system.

S.NO	SOFTWARE	PURPOSE
1	React.js	Frontend Development
2	Node.js	Backend Development
3	Express.js	API Handling
4	Supabase PostgreSQL	Database
5	Tailwind CSS	UI Design
6	VS Code	Development Environment
7	Chrome Browser	Testing

Tabel 3.2 Software Requirement Table

CHAPTER IV

SYSTEM ARCHITECTURE

4.1 Existing System

The existing voting system primarily depends on traditional paper ballots or Electronic Voting Machines (EVMs) deployed at physical polling stations. In this approach, voters are required to travel to designated polling locations where their identities are verified against official voter lists before they are allowed to cast their votes. Although this method has been widely used for many years, it presents several operational and administrative challenges.

Managing elections through physical infrastructure requires extensive planning, transportation of voting equipment, deployment of election personnel, and continuous security monitoring. These activities increase election costs and consume significant time and resources. The system also creates accessibility issues for citizens living in remote areas, migrant workers, elderly individuals, and people with disabilities, as they may find it difficult to reach polling stations.

Furthermore, physical voting records are susceptible to damage, loss, or tampering during transportation and storage. Manual verification and vote-counting procedures may introduce delays and human errors, affecting the efficiency of election management. The lack of remote voting capabilities further limits voter participation. As a result, traditional voting systems face challenges related to accessibility, scalability, security, and operational efficiency in modern democratic environments.

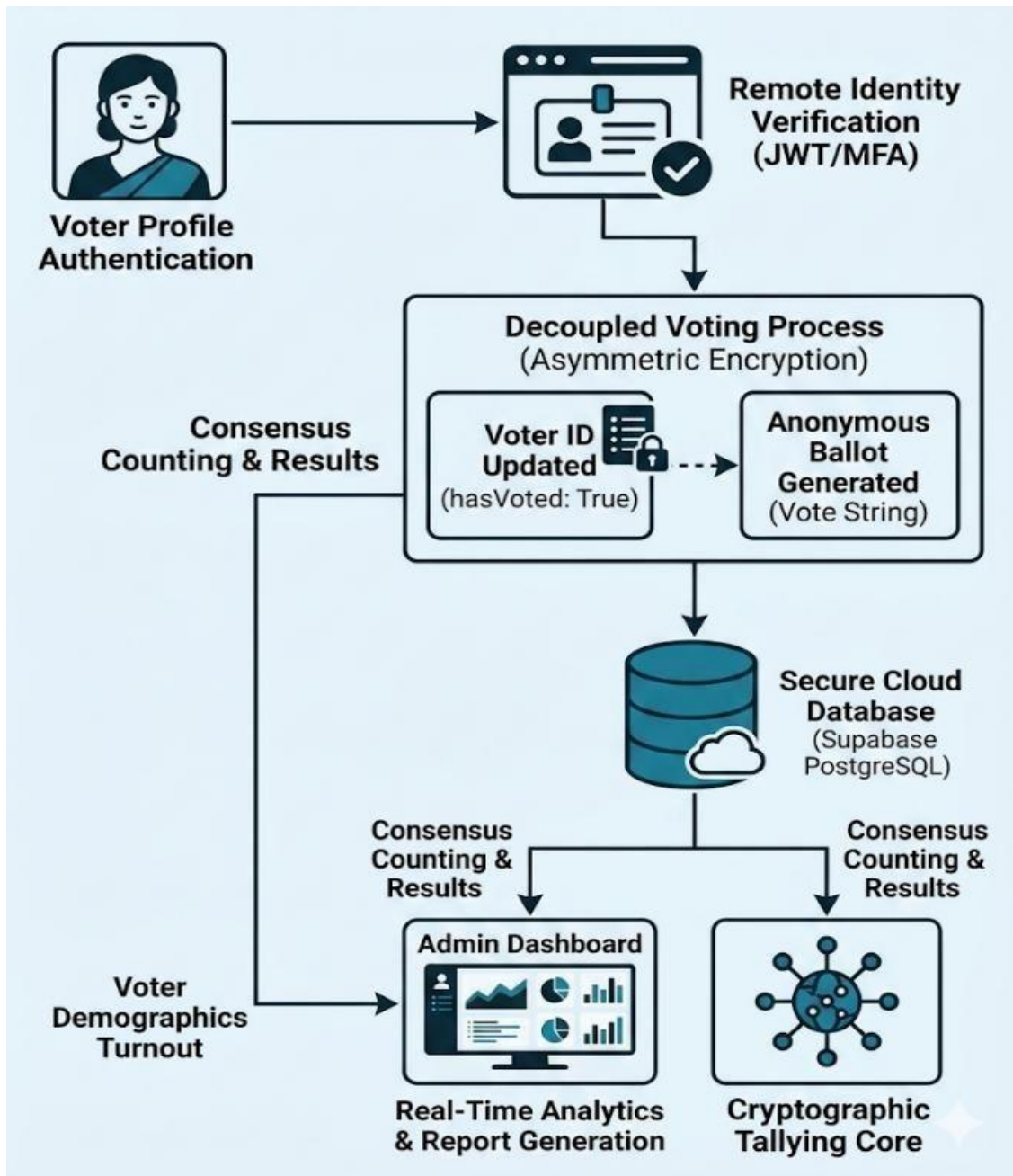


Figure 4.1 Existing System

4.2 Proposed System Architecture

The proposed Online Voting System introduces a secure cloud-based digital platform for conducting elections efficiently and transparently. The architecture follows a four-layer design consisting of the Frontend Layer, Backend Layer, Security Layer, and Database Layer.

Frontend Layer

The frontend is developed using React.js and Tailwind CSS. It provides a user-friendly interface for voters and election administrators. Voters can securely log in, view election details, and cast their votes online. Administrators can create elections, manage candidate information, monitor voting activities, and publish results.

Backend Layer

The backend is developed using Node.js and Express.js. It handles user authentication, vote processing, election management, and API communication. The backend ensures smooth interaction between the user interface and the database while enforcing election rules and access controls.

Security Layer

A dedicated security layer is integrated to protect voter privacy and election integrity. JWT-based authentication, encrypted communication channels, and secure session management are used to prevent unauthorized access. The system ensures that each eligible voter can cast only one vote while maintaining ballot confidentiality.

Database Layer

Supabase PostgreSQL is used as the cloud database for secure data storage. The database maintains voter information, candidate records, election details, voting history, and result data. Proper access controls ensure data consistency and protection against unauthorized modifications.

The proposed architecture improves accessibility, enhances security, enables real-time vote management, and significantly reduces the administrative workload associated with traditional election systems.

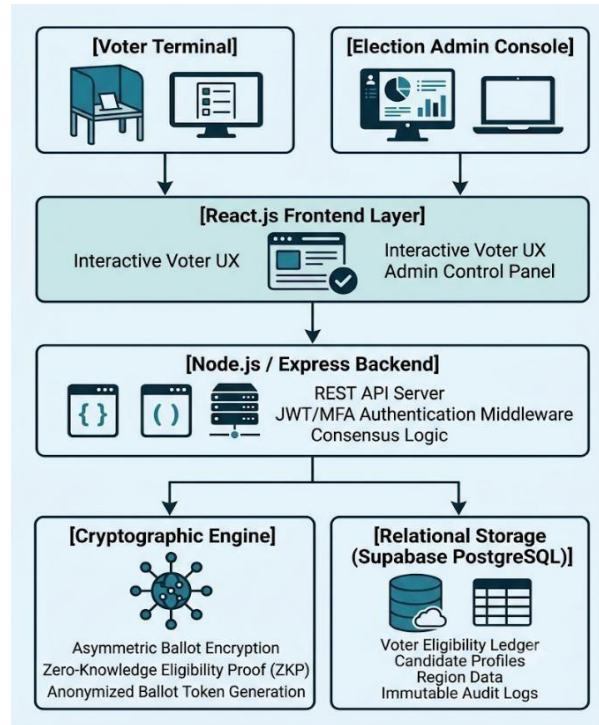


Figure 4.2 Proposed System Architecture

4.3 Novelty and Innovation Idea

The Online Voting System introduces several innovative features compared to traditional paper-based voting methods and conventional Electronic Voting Machine (EVM) systems. The system is designed to enhance accessibility, security, transparency, and efficiency in the electoral process.

One of the key innovations is the implementation of **Role-Based Access Control (RBAC)**, where voters and administrators access the system through separate authenticated portals. Voters can securely cast their votes and view election information, while administrators can manage elections, candidates, and results through dedicated dashboards. This ensures proper access control and data security.

Another innovative feature is the integration of **secure authentication and encrypted voting mechanisms**. JWT-based authentication, encrypted communication channels, and vote validation procedures help prevent unauthorized access, voter impersonation, and duplicate voting. The system guarantees that each eligible voter can cast only one vote while maintaining ballot secrecy.

The project also incorporates **real-time vote management and automated result generation**. Unlike traditional systems that require manual counting and verification, votes are recorded instantly in a secure cloud database, enabling faster and more accurate result processing.

Cloud-based storage using **Supabase PostgreSQL** provides reliable and secure management of voter,

The project also incorporates **real-time vote management** and **automated result generation**, significantly improving the efficiency of the electoral process. Unlike traditional voting systems that depend on manual counting and verification procedures, votes are recorded instantly in a secure cloud database. This enables faster vote processing, reduces human errors, and ensures accurate and timely declaration of election results.

Another important innovation is the use of **cloud-based data storage** through Supabase PostgreSQL. The database provides secure, reliable, and scalable management of voter information, candidate details, election records, and voting data. Since the system is hosted on the cloud, authorized users can securely access the platform from different locations, thereby improving accessibility for remote voters and election administrators.

The system further enhances transparency by maintaining digital records of voting activities while preserving voter privacy and ballot confidentiality. Secure authentication mechanisms, encrypted communication channels, and role-based access control ensure that only authorized users can access system resources.

These innovative features make the Online Voting System a secure, scalable, transparent, and user-friendly platform for conducting modern digital elections. The proposed solution minimizes administrative workload, improves voter participation, and supports efficient election management while maintaining the integrity and reliability of the democratic process.

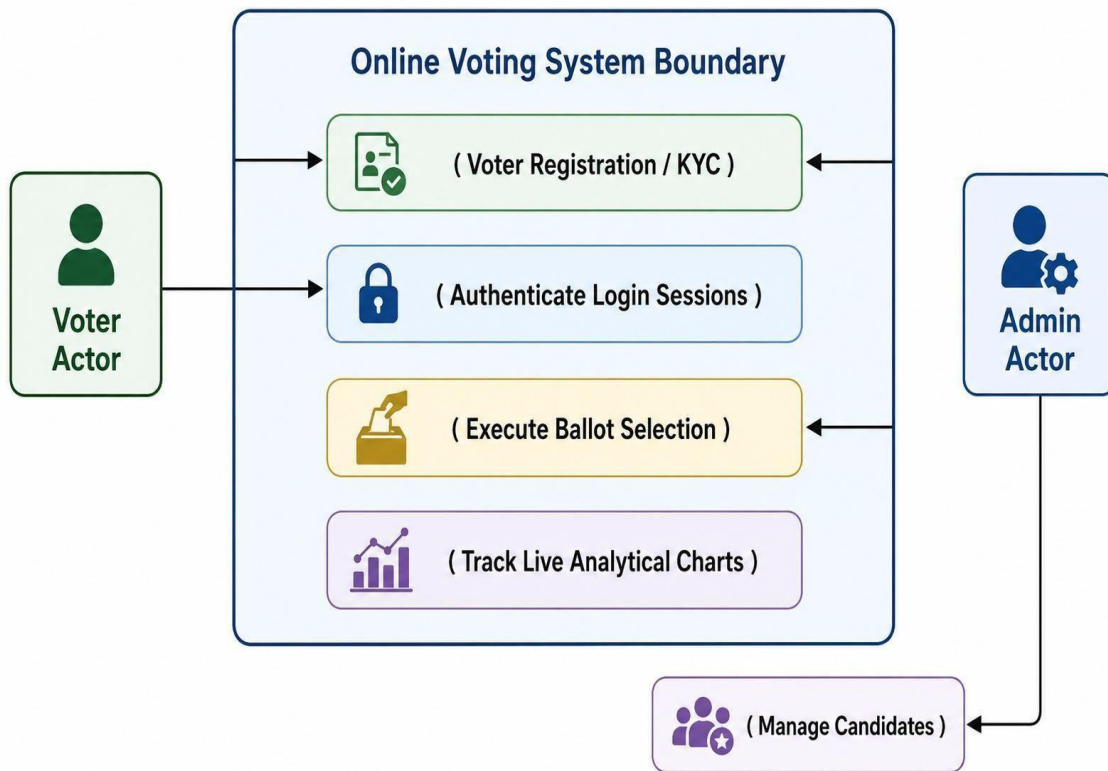


Figure 4.3 Novelty and Innovation Idea

CHAPTER V

HIGH LEVEL SYSTEM DESIGN

Introduction

The High-Level System Design describes the overall architecture and operational workflow of the Online Voting System. It explains how various system components interact with each other to provide a secure, transparent, and efficient digital voting environment. The system is designed using a multi-layer architecture consisting of the frontend layer, backend layer, security layer, and cloud database layer.

This chapter presents the structural design of the system and illustrates how data flows between voters, administrators, application services, and the database. The frontend layer provides user interfaces for voters and administrators, while the backend layer manages authentication, election processes, vote validation, and result generation. The security layer ensures voter privacy and election integrity through authentication and encryption mechanisms. The cloud database securely stores voter information, candidate details, election records, and voting data.

To provide a clear understanding of the system architecture and functionality, this chapter includes various UML diagrams and Data Flow Diagrams. These diagrams visually represent user interactions, system processes, database relationships, and communication between different modules. The diagrams included in this chapter are Use Case Diagram, Class Diagram, Sequence Diagram, Activity Diagram, and Data Flow Diagrams (DFD Level 0, Level 1, and Level 2).

These design models help in understanding the overall system structure, simplify the implementation process, and provide a comprehensive view of how the proposed Online Voting System operates in a secure and scalable manner.

5.1 Use Case Diagram

The Use Case Diagram represents the interaction between users and the Online Voting System. The primary users of the system are **Voters** and **Administrators**.

Administrators are responsible for managing elections, registering candidates, monitoring voting activities, verifying voter records, and publishing election results. They have complete control over election-related operations and system management.

Voters can securely log in to the system, view election details, access candidate information, cast their votes, and view election announcements. The system ensures that each eligible voter can vote only once while maintaining ballot confidentiality and security.

Functions of Administrator

- Manage voter records
- Register and manage candidates
- Create and manage elections
- Verify voter eligibility
- Monitor voting activities
- View voting statistics and reports
- Generate and publish election results
- Manage system settings and security

Functions of Voter

- Register/Login to the system
- View election information
- View candidate details
- Cast vote securely
- Verify voting status
- View election notifications
- Update profile information
- Logout securely

The Use Case Diagram provides a clear overview of user roles and system functionalities, helping stakeholders understand how the Online Voting System operates and how different users interact with the platform.

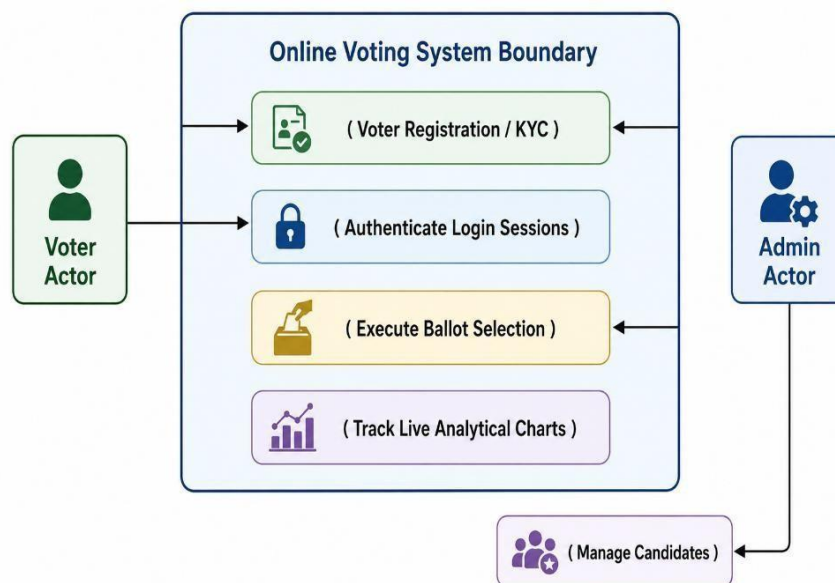


Figure 5.1 Use Case Diagram

5.2 Class Diagram

The Class Diagram represents the structural design of the Online Voting System and illustrates the relationships between various entities within the system. It defines the data objects, their attributes, and the interactions among them. The diagram helps in understanding how information is organized and managed throughout the voting process.

The major classes included in the system are:

- User
- Voter
- Administrator
- Election
- Candidate
- Vote
- Result

User

The User class stores common information such as user ID, name, email, password, and role. It serves as the base entity for both voters and administrators.

Voter

The Voter class contains voter-specific details such as voter ID, eligibility status, and voting status. It is responsible for voter authentication and participation in elections.

Administrator

The Administrator class manages election-related activities, including voter verification, candidate management, election creation, and result publication.

Election

The Election class stores information related to elections, such as election ID, title, description, start date, end date, and status.

Candidate

The Candidate class maintains candidate details, including candidate ID, name, party affiliation, symbol, and election association.

Result

The Result class manages vote counting and result generation. It stores candidate-wise vote totals and election outcomes.

The Class Diagram provides a clear representation of the database structure and relationships among users, elections, candidates, votes, and results, ensuring efficient management of the Online Voting System.

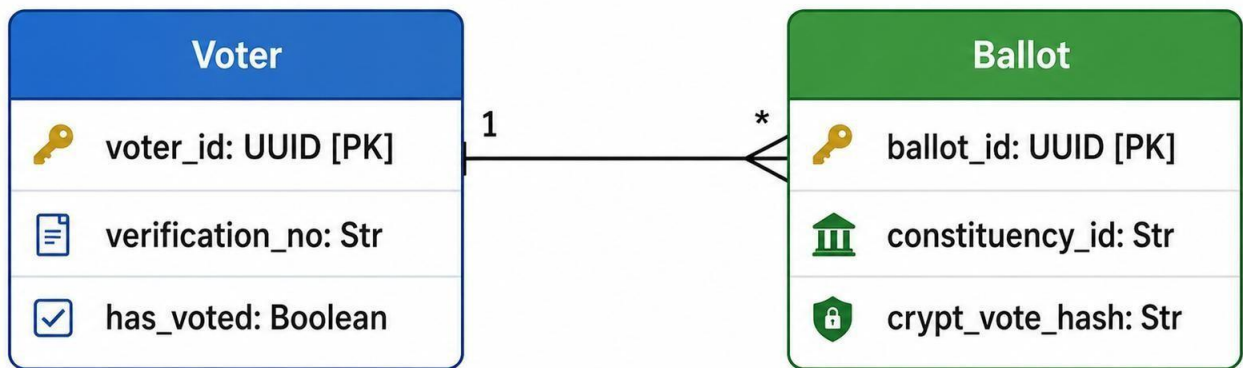


Figure 5.2 Class Diagram

The User class stores authentication details and role information for voters and administrators. The Voter class maintains voter-specific information, while the Administrator class manages elections and candidate records. The Election class stores election details, and the Candidate class contains candidate information. The Vote class records voting transactions securely, and the Result class manages vote counting and election outcomes. Relationships between these classes are maintained using primary keys and foreign keys to ensure proper database normalization, data integrity, and secure management of election data.

5.3 Sequence Diagram

The Sequence Diagram illustrates the sequence of interactions between the frontend interface, backend APIs, security layer, and cloud database in the Online Voting System. It shows how data and requests are exchanged when users perform operations such as login, election management, and vote casting.

When a user logs into the system, the frontend sends authentication requests to the backend server. The backend verifies the credentials with the database and grants access based on the user's role. After successful authentication, voters can access election details and cast their votes securely.

When a vote is submitted, the frontend sends the voting data to the backend. The backend validates voter eligibility, checks whether the voter has already voted, records the vote securely in the database, and updates the voting status. Finally, the system confirms successful vote submission and displays the appropriate response to the user.

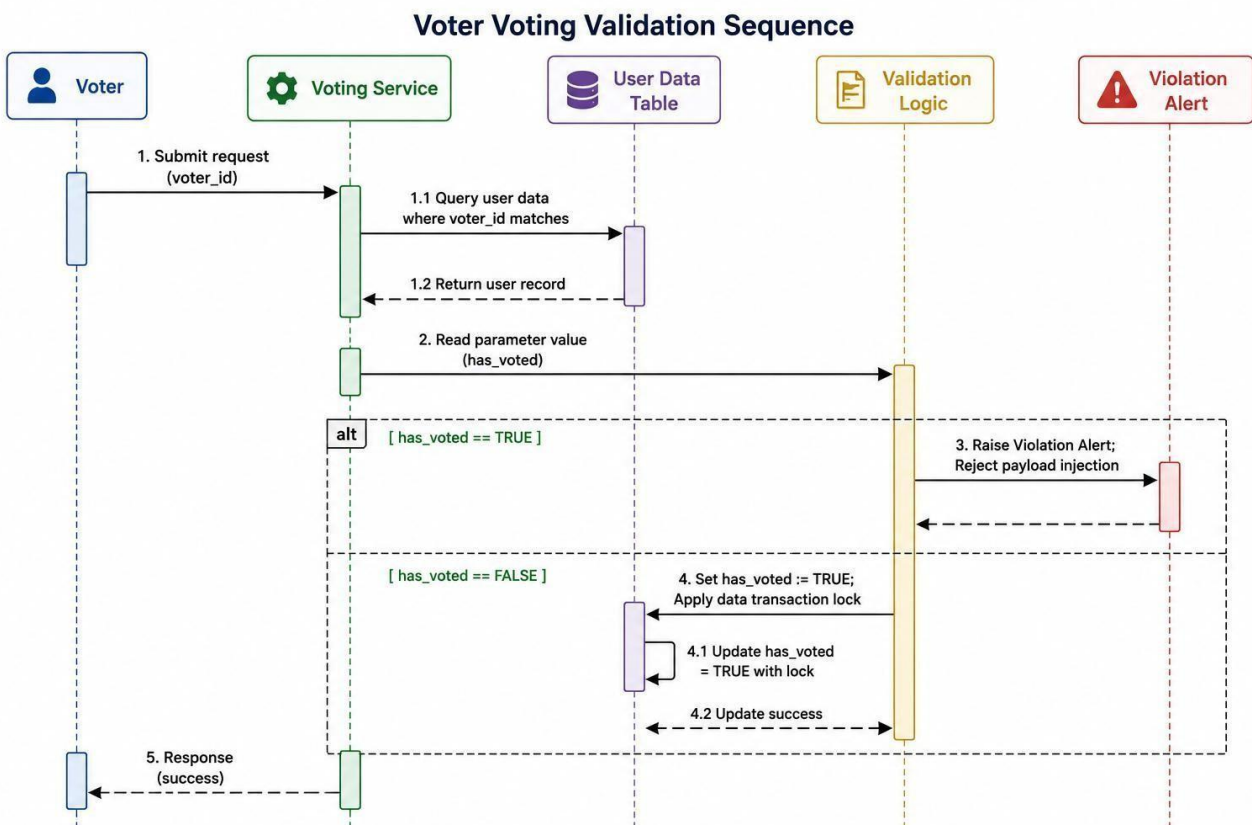


Figure 5.3 Sequence Diagram

5.4 Activity Diagram

The Activity Diagram represents the workflow of the Online Voting System. The process begins with user login, where the system verifies authentication credentials and determines the user's role. Based on the role, users are redirected to either the Voter Dashboard or Administrator Dashboard.

Administrators can create and manage elections, register candidates, verify voter eligibility, monitor voting activities, and publish election results. Voters can view election information, access candidate details, and cast their votes securely through the system.

The workflow also includes voter authentication, vote validation, secure vote recording, and automated result generation processes. After a vote is successfully submitted, the system updates the voting status, stores the vote securely in the database, and prevents duplicate voting. Finally, election results are generated automatically after the voting period ends, ensuring transparency, accuracy, and efficiency in the electoral process.

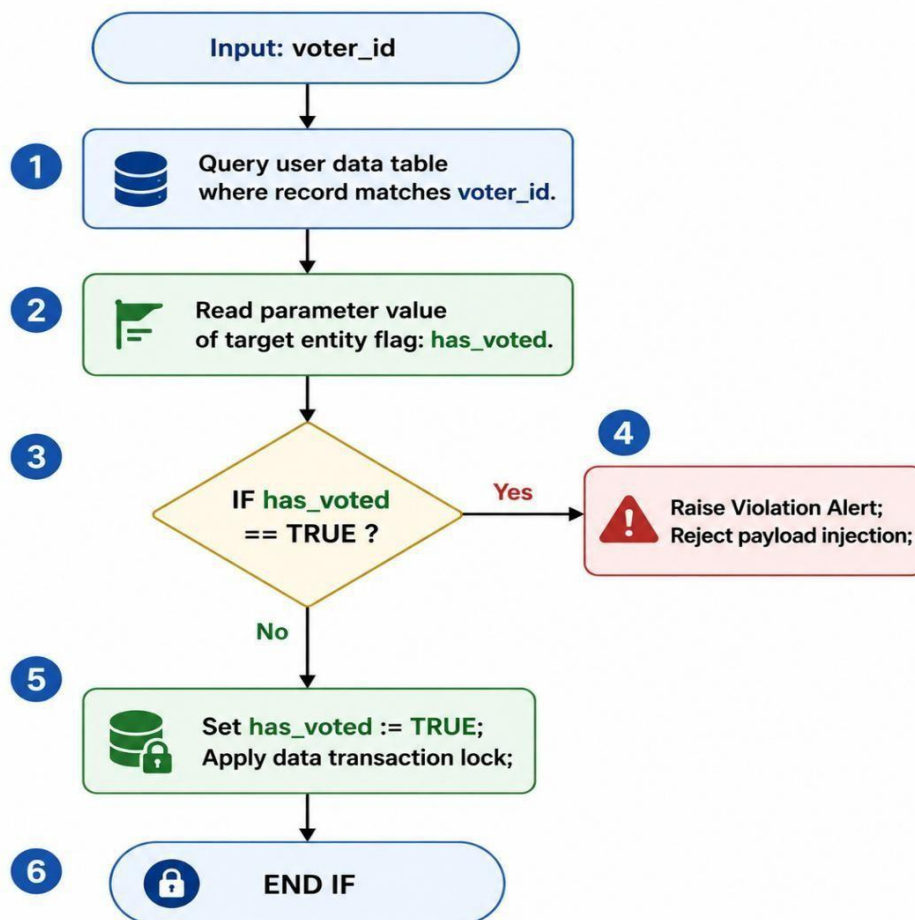


Figure 5.4 Activity Diagram

5.5 Level 0 Data Flow Diagram

The Level 0 Data Flow Diagram (DFD) provides a high-level overview of the entire Online Voting System. It illustrates the interaction between the primary users, namely Voters and Administrators, and the central voting application.

The diagram shows how voters access the system to authenticate themselves, view election information, and cast votes securely. Administrators interact with the system to manage elections, verify voters, register candidates, and publish election results. All election-related information, voter records, candidate details, voting data, and result information are processed through the system and stored in the cloud database.

The Level 0 DFD provides a simplified view of the overall data flow, helping users understand how information moves between external entities, the Online Voting System, and the database.

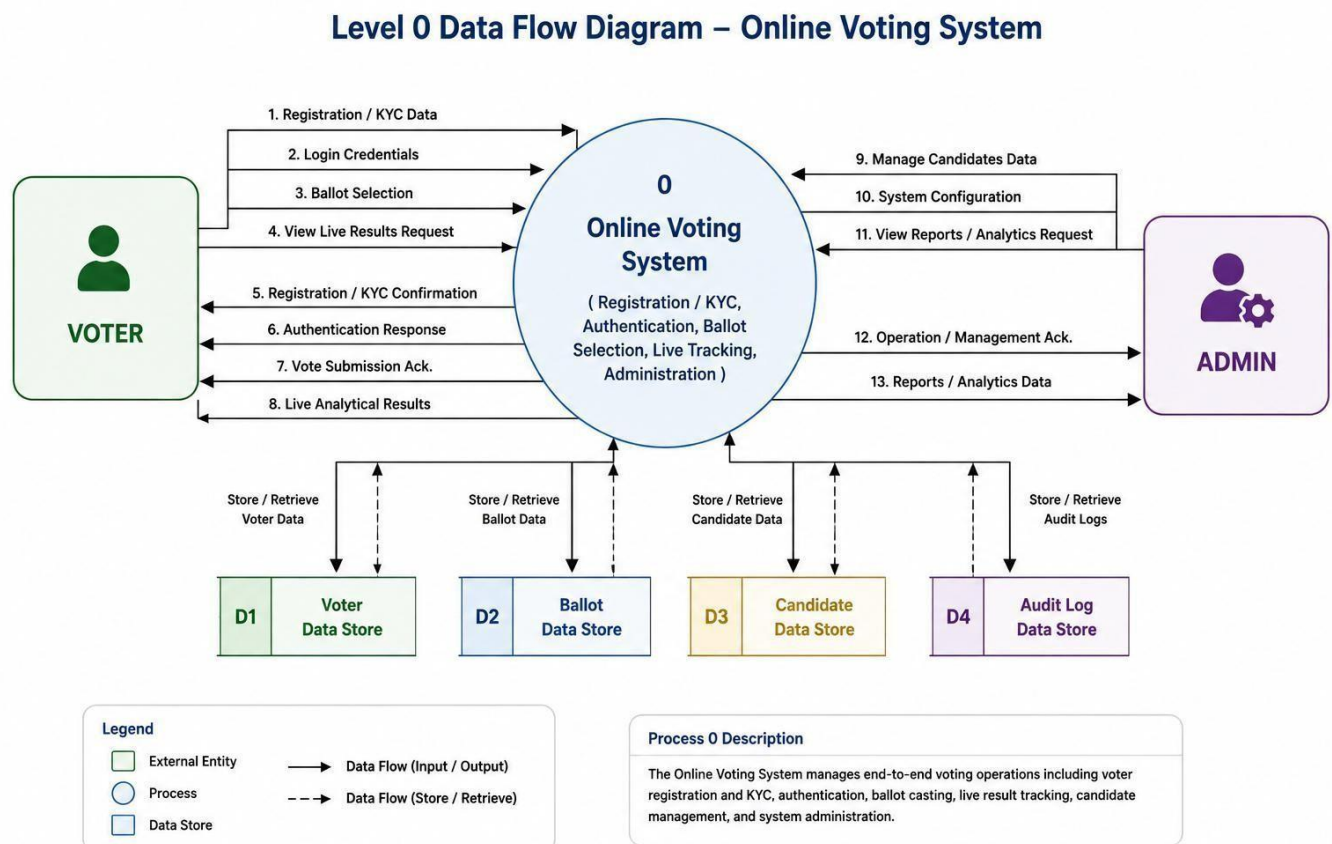


Figure 5.5 Level 0 Data Flow Diagram

5.6 Data Flow Diagram – Level 1

The Level 1 Data Flow Diagram (DFD) explains the major internal modules of the Online Voting System and how data flows between them. The system consists of the following key modules:

- Authentication Module
- Voter Management Module
- Candidate Management Module
- Election Management Module
- Vote Processing Module
- Result Generation Module

The Authentication Module verifies user credentials and grants access based on user roles. The Voter Management Module handles voter registration and eligibility verification. The Candidate Management Module manages candidate information and election participation. The Election Management Module is responsible for creating and managing elections.

The Vote Processing Module validates and records votes securely in the database, ensuring that each voter can vote only once. The Result Generation Module automatically counts votes and publishes election results. Each module processes user requests and exchanges data with the cloud database to ensure secure, accurate, and efficient election management.

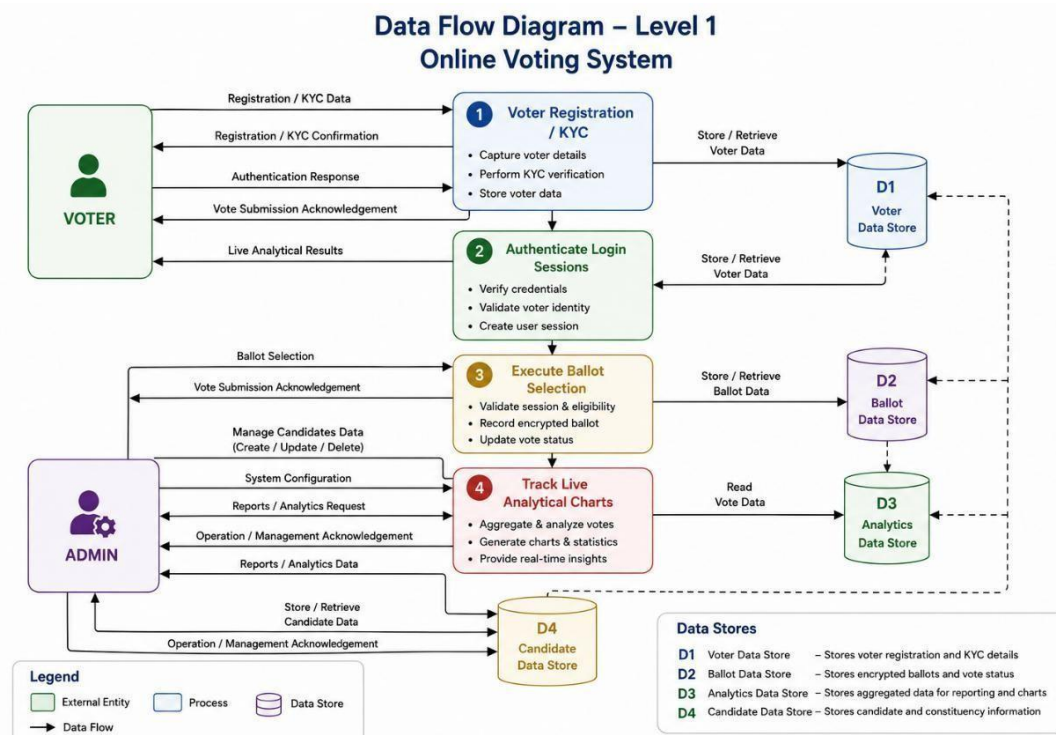


Figure 5.6 Data Flow Diagram – Level 1

5.7 Level 2 Data Flow Diagram

The Level 2 DFD provides a detailed view of the internal processing of the **Vote Processing Module** in the Online Voting System. The system receives voting requests from authenticated voters and performs a series of validation checks before recording the vote.

Initially, the system verifies voter identity and checks voter eligibility. It then confirms whether the voter has already cast a vote in the selected election. If the validation process is successful, the vote is securely recorded in the database, and the voter's voting status is updated.

The system then stores the voting information, updates election statistics, and prepares data for result generation. If any invalid or duplicate voting attempt is detected, the system rejects the request and displays an appropriate notification. This process ensures election integrity, prevents duplicate voting, and maintains secure and accurate vote management.

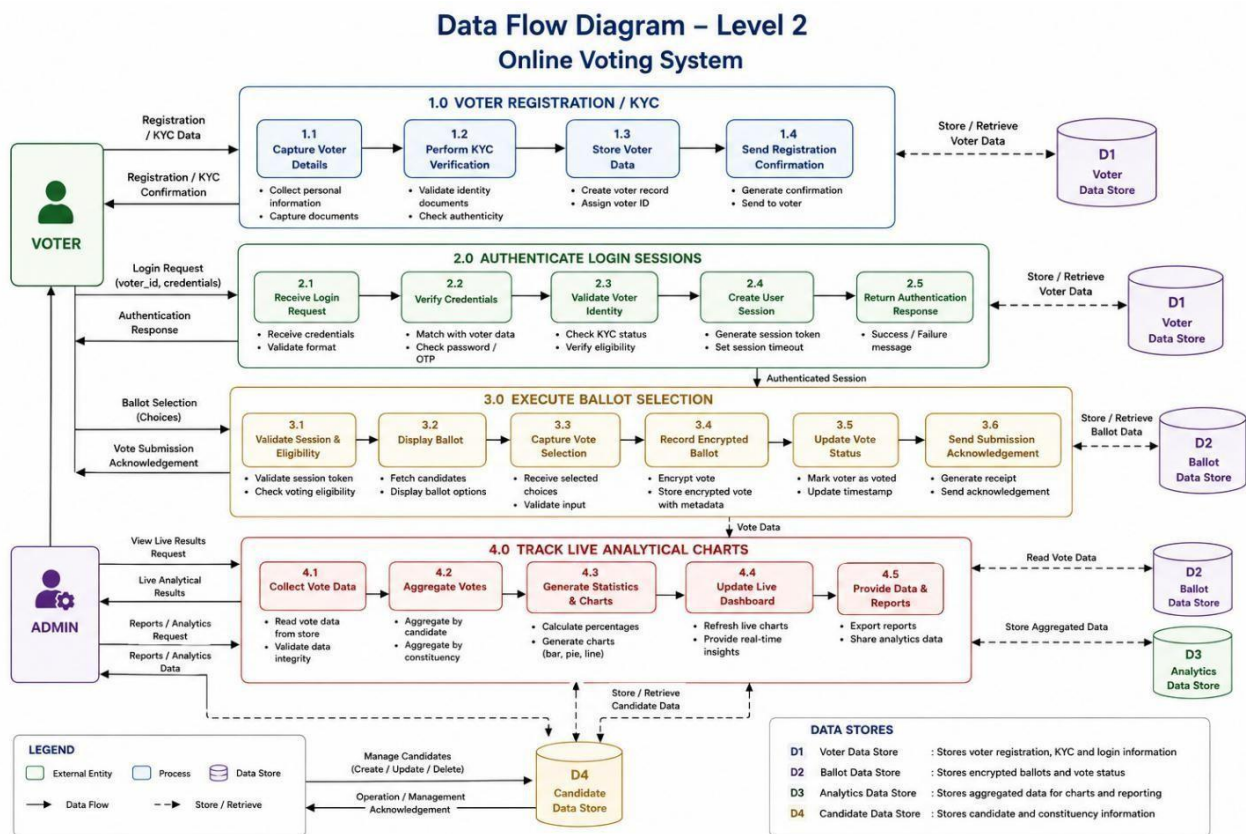


Figure 5.7 Level 2 Data Flow Diagram

CHAPTER VI

SYSTEM IMPLEMENTATION

6. Introduction

System Implementation is the process of transforming the designed architecture and system modules into a fully functional software application. The Online Voting System is implemented using modern full-stack web development technologies such as React.js, Node.js, Express.js, and Supabase PostgreSQL.

The implementation process includes frontend development, backend API integration, database management, user authentication, election management, vote processing, result generation, and security mechanisms. These components work together to provide a secure, efficient, and user-friendly digital voting platform.

The system is developed as a responsive web application that can be accessed through desktop computers, laptops, tablets, and mobile devices. Voters can securely log in, view election information, and cast votes online, while administrators can manage elections, candidates, voter records, and election results through dedicated dashboards. The implementation ensures secure communication, reliable data storage, and efficient handling of election-related operations while maintaining transparency, privacy, and scalability.

6.1 Assumptions

The following assumptions are considered during the development and implementation of the Online Voting System:

- Voters possess valid registration credentials and are eligible to participate in elections.
- Users have basic knowledge of using web applications and internet services.
- A stable internet connection is available while accessing the voting platform.
- Election and candidate information entered by administrators is accurate and verified.
- The cloud database server remains available, secure, and operational throughout the election process.
- Voters maintain the confidentiality of their login credentials.
- Administrators perform election management activities according to established election guidelines.

These assumptions help ensure the smooth, secure, and reliable operation of the proposed Online Voting System.

6.2 Algorithms

The Online Voting System utilizes several algorithms and mechanisms to ensure secure authentication, vote processing, election management, and result generation.

The major algorithms and processes implemented in the system are:

- User Authentication Algorithm
- JWT-Based Session Management
- Voter Eligibility Verification Algorithm

Secure Vote Recording Algorithm

- Duplicate Vote Prevention Mechanism
- Automated Result Generation Algorithm
- Database Management and Query Processing
- Role-Based Access Control (RBAC) System

These algorithms collectively ensure the security, accuracy, transparency, and efficiency of the online voting process

6.3 Voter Eligibility Verification Algorithm

Voter eligibility verification is one of the most important processes in the Online Voting System. The system automatically verifies voter credentials and eligibility before allowing a user to participate in an election. This process ensures that only authorized and registered voters can cast votes.

The algorithm validates the voter's identity using registered login credentials and checks whether the voter is eligible for the selected election. It also verifies that the voter has not already cast a vote in the same election. If all verification conditions are satisfied, the system grants access to the voting interface; otherwise, access is denied.

Algorithm Steps:

1. User enters login credentials.
2. System authenticates the user.
3. Verify voter registration and eligibility status.
4. Check whether the voter has already voted.
5. If eligible and not voted, allow vote casting.
6. Otherwise, display an appropriate error message.

This algorithm helps maintain election integrity, prevents unauthorized participation, and ensures that each eligible voter can cast only one vote.

CHAPTER VII

GUI / WORKING MODULES / EXPERIMENTAL RESULTS

Introduction

7.1 GUI / Working Modules

The Online Voting System is designed as a secure and user-friendly web application that enables voters and administrators to perform election-related activities efficiently. The graphical user interface (GUI) is developed using React.js and provides responsive access through desktop and mobile devices. The system consists of several functional modules that work together to ensure secure election management and vote processing.

7.1.1 User Authentication Module

The User Authentication Module is responsible for verifying the identity of users before granting access to the system. Voters and administrators log in using their registered credentials. The module validates user information through the backend server and database. JWT-based authentication is used to maintain secure sessions and prevent unauthorized access.

7.1.2 Voter Management Module

The Voter Management Module allows administrators to register and manage voter records. This module stores voter information, verifies eligibility, and maintains voting status. It ensures that only authorized and eligible voters can participate in elections. The module also helps administrators update voter information whenever necessary.

7.1.3 Voter Dashboard Module

The Voter Dashboard serves as the primary interface for voters. After successful login, voters can view active elections, candidate information, election schedules, and voting instructions. The dashboard provides a simple and intuitive interface that allows users to cast votes securely. Voters can also view their voting status and election notifications through this module.

7.1.4 Administrator Dashboard Module

The Administrator Dashboard provides comprehensive control over election operations. Administrators can create elections, manage candidates, verify voter records, monitor voting activities, and publish results. The dashboard also provides statistical information and reports that help administrators monitor the progress of elections in real time.

7.1.5 Election Management Module

The Election Management Module enables administrators to create and configure elections. Election details such as title, description, start date, end date, and candidate information are managed through this module. Administrators can activate or deactivate elections and ensure that election schedules are properly maintained.

7.1.6 Vote Processing Module

The Vote Processing Module is one of the most critical components of the system. When a voter submits a vote, the module verifies voter eligibility and checks whether the voter has already voted. Once validation is completed successfully, the vote is securely recorded in the database. The module prevents duplicate voting and ensures that ballot confidentiality is maintained throughout the process.

7.1.7 Result Generation Module

The Result Generation Module automatically calculates election results after the voting period has ended. The system counts votes stored in the database and generates accurate election outcomes without manual intervention. This automated process reduces counting errors and significantly improves result-processing efficiency.

7.1.8 Database Module

The Database Module uses Supabase PostgreSQL to store and manage voter records, candidate information, election data, voting history, and election results. The database provides secure storage, reliable data retrieval, and efficient management of large volumes of election-related information.

7.2 Result Analysis and Discussion

The Online Voting System was successfully implemented and evaluated under different operational scenarios. Functional testing confirmed that all modules operated according to the specified requirements. User authentication was performed securely, and role-based access control ensured that voters and administrators accessed only authorized functionalities.

The vote processing mechanism successfully validated voter eligibility and prevented duplicate voting attempts. All votes were accurately stored in the cloud database, maintaining data integrity and confidentiality. Election management functions enabled administrators to create elections, manage candidates, and monitor voting activities efficiently.

Performance testing demonstrated that the system could process voting requests quickly while maintaining stable operation. Automated result generation significantly reduced the time required for vote counting compared to traditional election methods. The cloud-based architecture improved accessibility by allowing users to participate from different locations using internet-enabled devices.

The experimental results indicate that the proposed Online Voting System provides a secure, scalable, transparent, and efficient platform for conducting digital elections. By reducing manual workload, enhancing voter accessibility, and ensuring accurate vote management, the system successfully achieves the objectives of modern electronic voting and contributes to the advancement of secure digital governance.

7.2.1 Result Generation Module

The Result Generation Module automatically calculates election results after the voting period has ended. The system counts votes stored in the database and generates accurate election outcomes without manual intervention. This automated process reduces counting errors and significantly improves result-processing efficiency.

7.2.2 Database Module

The Database Module uses Supabase PostgreSQL to store and manage voter records, candidate information, election data, voting history, and election results. The database provides secure storage, reliable data retrieval, and efficient management of large volumes of election-related information.

7.3 Result Analysis and Discussion

The Online Voting System was successfully implemented and evaluated under different operational scenarios. Functional testing confirmed that all modules operated according to the specified requirements. User authentication was performed securely, and role-based access control ensured that voters and administrators accessed only authorized functionalities.

The vote processing mechanism successfully validated voter eligibility and prevented duplicate voting attempts. All votes were accurately stored in the cloud database, maintaining data integrity and confidentiality. Election management functions enabled administrators to create elections, manage candidates, and monitor voting activities efficiently.

Performance testing demonstrated that the system could process voting requests quickly while maintaining stable operation. Automated result generation significantly reduced the time required for vote counting compared to traditional election methods. The cloud-based architecture improved accessibility by allowing users to participate from different locations using internet-enabled devices.

The experimental results indicate that the proposed Online Voting System provides a secure, scalable, transparent, and efficient platform for conducting digital elections. By reducing manual workload, enhancing voter accessibility, and ensuring accurate vote management, the system successfully achieves the objectives of modern electronic voting and contributes to the advancement of secure digital governance.

CHAPTER VIII

TESTING

TESTING

Testing is an important phase in software development that ensures the Online Voting System functions correctly, securely, and efficiently. The primary objective of testing is to identify defects, verify system requirements, and ensure that all modules perform as expected. Various testing techniques were applied to validate user authentication, election management, vote processing, result generation, and database operations. The testing process helped ensure system reliability, security, and accuracy before deployment.

8.1 Test Plan

The test plan defines the strategy used to evaluate the Online Voting System. The main objectives of testing are to verify functionality, validate security mechanisms, ensure data integrity, and confirm system performance under different conditions.

The following modules were included in the testing process:

- User Authentication Module

- Voter Management Module

- Election Management Module

- Candidate Management Module

- Vote Processing Module

- Result Generation Module

- Database Management Module

The testing activities included functional testing, integration testing, security testing, and user interface testing. Each module was tested independently and then evaluated as part of the complete system to ensure proper interaction between components.

8.2 Test Cases

The following test cases were executed to validate system functionality:

Test Case ID	Test Description	Expected Result
TC01	Login with valid credentials	User successfully logs into the system
TC02	Login with invalid credentials	Error message displayed and access denied
TC03	Create a new election	Election created successfully
TC04	Add candidate to election	Candidate information stored successfully
TC05	Verify eligible voter	Voter allowed to access voting page
TC06	Cast a vote successfully	Vote recorded in database
TC07	Attempt duplicate voting	System rejects duplicate vote
TC08	View election results	Results displayed correctly
TC09	Generate automated results	Vote count calculated accurately
TC10	Logout from system	User session terminated successfully

Table8.2 Test Cases

These test cases cover the major functionalities of the Online Voting System and ensure that all critical operations perform correctly.

8.3 Test Results

The testing results demonstrated that the Online Voting System met all functional and security requirements. Authentication mechanisms successfully validated user identities and prevented unauthorized access. Election management operations such as election creation, candidate registration, and voter verification worked correctly without errors.

The vote processing module accurately recorded votes and effectively prevented duplicate voting attempts. Automated result generation produced correct election outcomes with minimal processing time. Database operations were executed successfully, ensuring reliable storage and retrieval of election data.

Overall, the system achieved a high level of accuracy, security, and reliability during testing. The successful completion of all test cases confirms that the Online Voting System is suitable for conducting secure and efficient digital elections.

CHAPTER IX

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The Online Voting System was successfully designed and implemented to provide a secure, transparent, and efficient digital platform for conducting elections. The system addresses many limitations of traditional voting methods, including geographical restrictions, manual vote counting, and administrative complexity. By utilizing modern web technologies such as React.js, Node.js, Express.js, and Supabase PostgreSQL, the platform delivers a reliable and user-friendly voting experience.

The system incorporates secure authentication, role-based access control, vote validation mechanisms, and cloud-based data management to ensure election integrity and voter privacy. Voters can participate in elections remotely through internet-enabled devices, while administrators can efficiently manage elections, candidates, voter records, and results through dedicated dashboards.

The implementation and testing results demonstrate that the system successfully records votes, prevents duplicate voting, generates accurate results, and maintains secure election data. Automated vote processing and result generation significantly reduce the time and effort required compared to traditional election procedures. Overall, the Online Voting System provides a scalable, secure, and effective solution for modern digital elections and contributes to the advancement of technology-driven democratic processes.

9.2 Future Scope

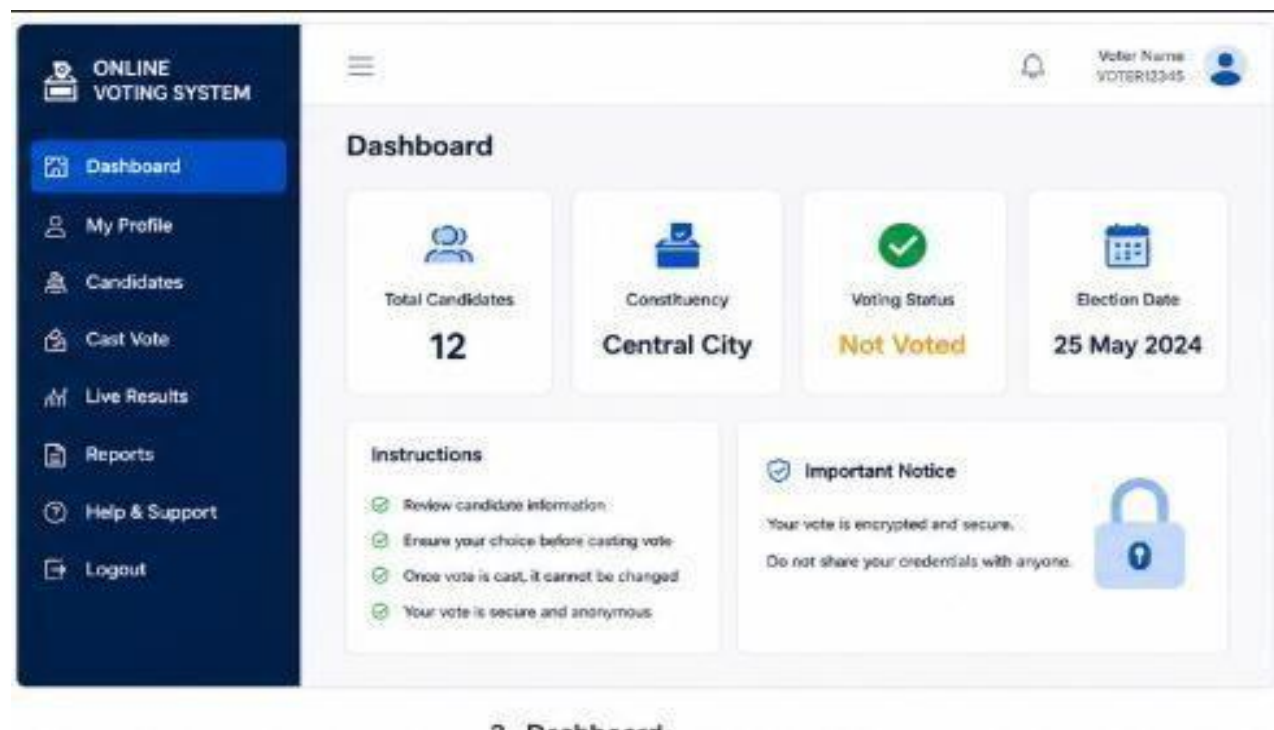
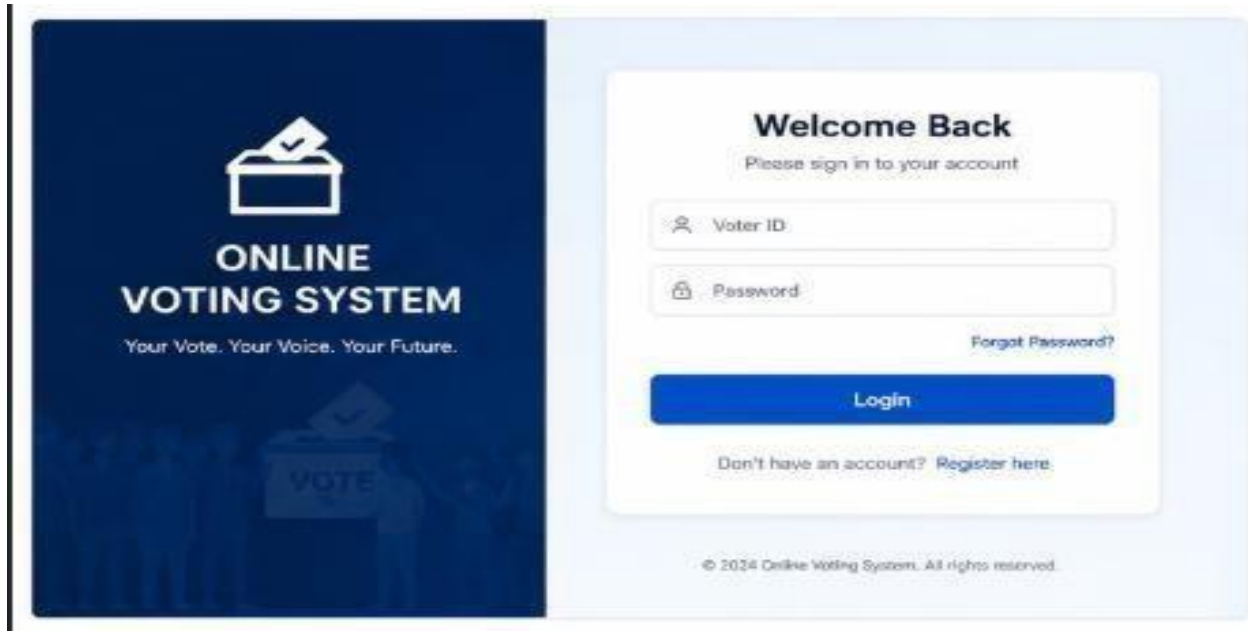
The Online Voting System can be further enhanced by integrating advanced technologies and additional security features. Future versions of the system may incorporate biometric authentication methods such as fingerprint or facial recognition to strengthen voter identity verification and reduce the risk of impersonation.

Blockchain technology can be integrated to provide immutable vote records and improve transparency, auditability, and public trust in election outcomes. Mobile application support can also be developed to increase accessibility and provide a more convenient voting experience for users.

Additional enhancements may include multilingual interfaces, real-time election analytics, advanced reporting dashboards, AI-based fraud detection mechanisms, and support for large-scale national elections. Integration with government identity verification systems can further improve voter authentication and election security.

With continuous technological improvements, the Online Voting System has the potential to evolve into a highly secure, scalable, and globally adaptable platform capable of supporting future digital governance and electronic election initiatives.

OUTPUT



ONLINE VOTING SYSTEM

Dashboard

My Profile

Candidates

Cast Vote

Live Results

Reports

Help & Support

Logout

Cast Your Vote

Select a candidate of your choice

Constituency: Central City

Election Date: 25 May 2024



Arjun Patel

Independent



Meera Sharma

Progressive Party



Rohan Verma

People's Party



Sneha Iyer

Unity Alliance

Please review your selection carefully.
Once you cast your vote, it cannot be changed.

Cancel

Review Vote

ONLINE VOTING SYSTEM

Dashboard

My Profile

Candidates

Cast Vote

Live Results

Reports

Help & Support

Logout

Confirm Your Vote

Please review your selection before final submission



Arjun Patel

Independent

You have selected the following candidate:

Your vote is encrypted and secure.
It cannot be modified once submitted.

Back

Confirm & Submit Vote

34

OUTPUT EXPLANATION

1. Login Page Output

The Login Page allows voters and administrators to authenticate themselves using their registered credentials. Upon successful verification, the system redirects users to their respective dashboards based on their assigned roles. Invalid login attempts are rejected with appropriate error messages to maintain system security.

2. Voter Dashboard Output

The Voter Dashboard displays election information, candidate details, voting status, and election notifications. Users can access active elections and securely participate in the voting process through this interface. The dashboard provides a simple and user-friendly voting experience.

3. Administrator Dashboard Output

The Administrator Dashboard provides complete control over election management activities. Administrators can create elections, register candidates, verify voter records, monitor voting progress, and view election statistics. The dashboard serves as the central control panel for managing the entire voting system.

4. Election Management Output

The Election Management Module allows administrators to create and configure elections. Election details such as election name, schedule, and status are displayed and managed through this interface. Successful election creation confirms that the system is ready to accept votes during the election period.

5. Candidate Management Output

The Candidate Management Module enables administrators to add, update, and manage candidate information. Candidate records are stored securely in the database and displayed to voters during the voting process.

6. Vote Casting Output

The Vote Casting Module allows eligible voters to select their preferred candidate and submit their vote securely. The system validates voter eligibility and records the vote in the database. A confirmation message is displayed after successful vote submission.

7. Vote Validation Output

The Vote Validation Module checks whether the voter is authorized and verifies that the voter has not already participated in the election. Duplicate voting attempts are automatically blocked, ensuring election integrity.

8. Result Generation Output

The Result Generation Module automatically counts all valid votes stored in the database. The system calculates vote totals for each candidate and generates election results without manual intervention. This reduces errors and improves efficiency.

9. Election Results Output

The Election Results Page displays candidate-wise vote counts and election outcomes. Results are presented clearly and accurately, allowing administrators and authorized users to analyze election performance.

APPENDICES

Appendix 1 – Sample Screenshots of Online Voting System

This appendix contains sample screenshots of the developed Online Voting System interface. The screenshots demonstrate the functionality and user interaction of various modules implemented in the system.

The screenshots may include:

- Login Page
- Voter Registration Page
- Voter Dashboard
- Administrator Dashboard
- Election Management Page
- Candidate Management Page
- Voting Interface
- Vote Confirmation Page
- Result Generation Page
- Election Results Dashboard

These screenshots provide visual evidence of the successful implementation of the proposed system and help in understanding the workflow of different modules.

Appendix 2 – Sample Database Tables and Source Code Snippets

This appendix contains sample database structures and important source code snippets used in the implementation of the Online Voting System.

The appendix may include:

- User Table Structure
- Voter Table Structure
- Candidate Table Structure
- Election Table Structure
- Vote Table Structure
- Result Table Structure
- Authentication Module Code
- Vote Validation Logic
- Result Generation Queries
- Database Connection Configuration

These technical details provide additional information regarding the internal implementation of the system.

REFERENCES

1. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education, 2016.
2. Roger S. Pressman and Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th Edition, McGraw-Hill Education, 2020.
3. Efraim Turban, Linda Volonino, and Gregory Wood, *Information Technology for Management*, Wiley Publications, 2018.
4. Abraham Silberschatz, Henry F. Korth, and S. Sudarshan, *Database System Concepts*, 7th Edition, McGraw-Hill Education, 2019.
5. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 4th Edition, MIT Press, 2022.
6. React.js Official Documentation. Available at: <https://react.dev/>
7. Node.js Official Documentation. Available at: <https://nodejs.org/>
8. Express.js Official Documentation. Available at: <https://expressjs.com/>
9. Supabase Official Documentation. Available at: <https://supabase.com/docs>
10. Mozilla Developer Network (MDN) Web Documentation. Available at: <https://developer.mozilla.org/>
11. OWASP Foundation. *Web Application Security Guidelines*. Available at: <https://owasp.org/>
12. Research papers and journals related to Electronic Voting Systems, Cyber Security, and Digital Governance.